Neural Network Potential Training on the ANI-1 Dataset

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Introduction

Predicting the energy of a molecule accurately is one of the main challenges in computational chemistry. The standard approach is to use quantum mechanical (QM) density functional theory (DFT), which can predict energies with high accuracy but at a high computational cost. This makes DFT an impractical method for large-scale molecular simulations. Classical force fields are significantly faster but have less accuracy, especially for non-equilibrium geometries and chemical reactions. On the other hand, neural network potentials (NNPs) can be trained on DFT data and then predict energies at a fraction of the computational cost, while having close to DFT accuracy.

This project implements and trains an ANI-style neural network potential on a subset of the ANI-1 dataset, which is the first truly transferable NNP for organic molecules, introduced by Smith, Isayev, and Roitberg in 2017[1]. The ANI-1 model uses atom-centered symmetry functions, originally developed by Behler and Parrinello[2] , to construct Atomic Environment Vectors (AEVs) as molecular representations and learns a separate neural network for each element type (H, C, N, O). The total molecular energy is calculated as the sum of atomic contributions from each element's network.

The dataset used in this work is the ANI-1 dataset, which contains DFT-calculated energies for small organic molecules drawn from the GDB-11 database. Molecules contain up to 8 heavy atoms (C, N, O) and hydrogen (H), while conformations were generated using Normal Mode Sampling (NMS). All models were implemented using PyTorch[3] and the TorchANI library[4]. In this project, the study trains on the GDB s01–s04 subset (molecules with 1–4 heavy atoms) rather than the full ANI-1 dataset and evaluates how closely the model's accuracy approaches the original paper's reported results.

Methods

The ANI-1 dataset was loaded using the TorchANI library[4]. Atomic self-energies were subtracted from all total energies to obtain relative conformational energies that are easier to learn. Species labels (H, C, N, O) were mapped to integer indices 0–3. The dataset was split into training (80%), validation (10%), and test (10%) sets, and batched using torchani's built-in collate and cache methods.

Each atom's local chemical environment is encoded as an AEV using modified Behler-Parrinello symmetry functions [12]. The AEV has a radial component (calculating distances to neighbors) and an angular component (calculating triplet angles), both within a cutoff radius. AEVs are differentiated by atom type where separate subvectors are computed for each neighbor element type. With 4 element types, this produces 4 radial and 10 angular subAEVs, giving a total dimensionality of 384. The parameters were radial cutoff 5.2 Å, angular cutoff 3.5 Å, 16 radial and 8 angular shifting values.

A separate atomic neural network is trained for each element type. Each network maps a 384-dimensional AEV to a scalar atomic energy. Two different base architectures were used:
AtomicNet_A: 384 → 128 → 1 (one hidden layer, ReLU)
AtomicNet_B: 384 → 256 → 128 → 1 (two hidden layers, ReLU)
All models were trained with the Adam optimizer (MSE loss) and early stopping based on validation loss, saving the best weights observed.

Five configurations were evaluated (Table 1), varying architecture, learning rate, epochs, L2 weight decay, and batch size. Each model was initialized independently before training for fair comparison. Model 3 (AtomicNet_B, lr=1e-3, epoch=20, l2=0, batch=8192) achieved the best test MAE of 1.24 kcal/mol, thus being selected as the final model.

Model	Architecture	LR	Epochs	L2	Batch	Dropout	Train MAE (kcal/mol)	Val MAE (kcal/mol)	Test MAE (kcal/mol)
1	384→128→1	1e-3	20	0	8192	0.0	1.74	1.75	1.76
2	384→128→1	1e-3	20	1e-4	8192	0.0	2.54	2.55	2.56
3	384→256→128→1	1e-3	20	0	8192	0.0	1.24	1.25	1.24
4	384→128→1	1e-4	20	1e-5	8192	0.0	2.54	2.55	2.56
5	384→256→128→1	5e-4	30	1e-5	4096	0.0	1.54	1.54	1.55

Table 1: Hyperparameter search results. Test MAE = mean(|true − pred|) × 627.51 kcal/mol.

All error values use Mean Absolute Error (MAE), which is the mean of |true − predicted energy| across all molecules converted to kcal/mol via 627.5095 kcal/mol/hartree. The parity plot labels utilize the same formula. The ANI-1 paper uses the Root Mean Squared Error (RMSE), where RMSE ≥ MAE, so the numerical comparisons between the two metrics are approximate.

The study used two experiments to assess the robustness of the best model. One was five independent training runs from different random initialization and the other was 5-fold cross validation on the combined train and validation set. Finally, a production model was trained on all available data (train, validation, and test) to maximize usage.

Results

Table 1 summarizes all five configurations of hyperparameters. AtomicNet_B consistently outperformed AtomicNet_A across all metrics. Model 3 (AtomicNet_B, lr=1e-3, l2=0) achieved the best test MAE of 1.24 kcal/mol and is the only model below the 2 kcal/mol target. The train, validation, and test MAEs for Model 3 are nearly identical (1.24, 1.25, 1.24 kcal/mol), showing no evidence of overfitting. Adding L2 regularization (Model 2) hurt performance, making test MAE 2.56 kcal/mol, suggesting the dataset is not large enough for regularization to help. Model

5 (also AtomicNet_B but lower lr and smaller batch) performed worse at 1.55 kcal/mol, suggesting lr=1e-3 with batch=8192 provides more effective convergence within 20 epochs. No dropout was used across all models.

Five independent runs of Model 3 produced test MAEs of 1.48, 1.92, 2.00, 1.57, and 2.48 kcal/mol, with a mean of 1.890 ± 0.353 kcal/mol. The variance reflects sensitivity to random initialization on a smaller dataset. The best single run (1.24 kcal/mol from the hyperparameter search) being lower than this mean suggests that this run had a favorable initialization relative to a typical outcome.

5-fold cross validation yielded test MAEs of 1.10, 1.30, 1.18, 1.38, and 1.72 kcal/mol per fold with a mean of 1.336 ± 0.213 kcal/mol. The closeness between validation and test MAE across all folds (val mean: 1.338 ± 0.216) confirms the consistent generalization. Figure 1 shows the CV bar chart.

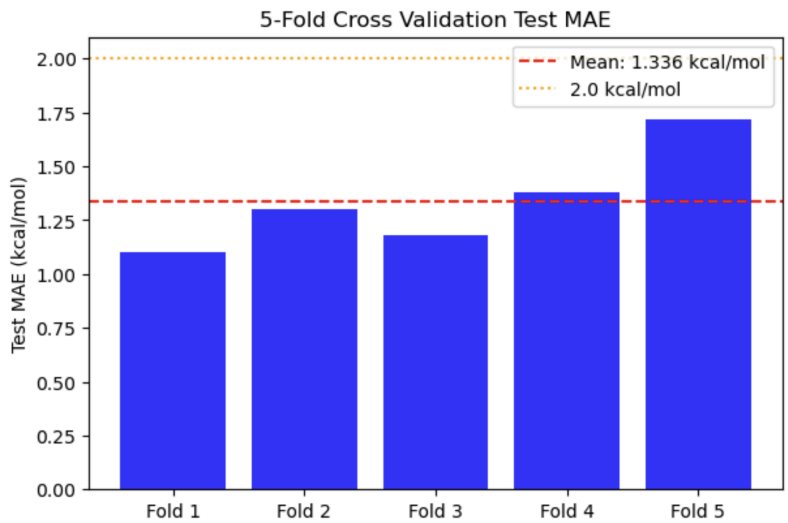

Figure 1: 5-Fold cross-validation test MAE per fold. Red dashed line = mean (1.336 kcal/mol). Orange dotted line = 2.0 kcal/mol target.

The final model trained on all available data achieved a MAE of 1.11 kcal/mol. The learning curve (Figure 2) shows smooth convergence of both train and validation loss over 20 epochs with no overfitting. The parity plot (Figure 3) shows tight clustering around the y=x diagonal across the full energy range, with no systematic bias.

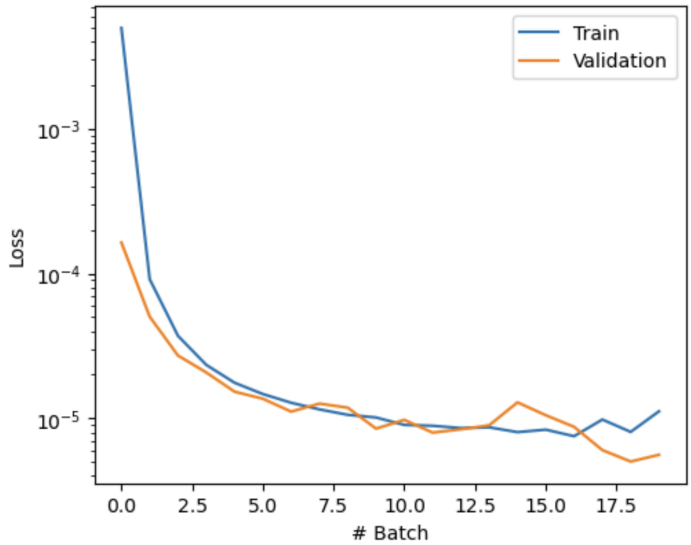

Figure 2: Training and validation loss curves for the final model over 20 epochs.

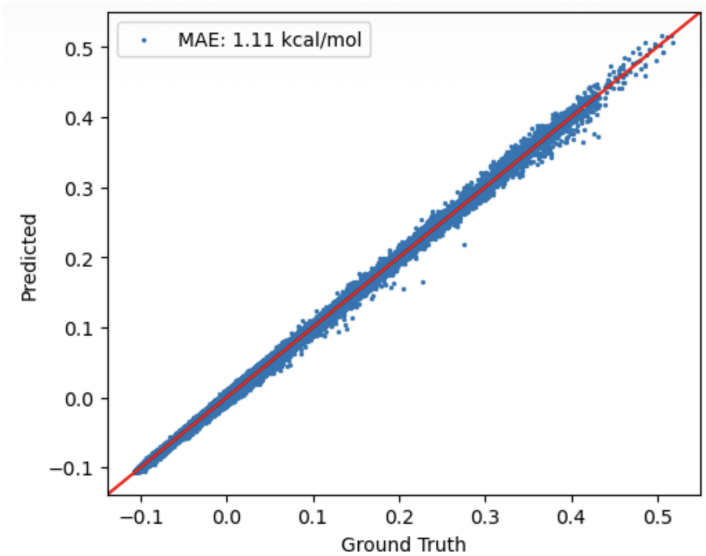

Figure 3: Parity plot for the final model trained on all data. Points are tightly distributed on the y=x diagonal.

Table 2 compares our results to Smith et al. (2017). Our 5-fold CV mean of 1.336 kcal/mol (MAE) is similar to the paper's test RMSE of 1.3 kcal/mol, even considering the fact that RMSE ≥ MAE. Our final model achieves 1.11 kcal/mol MAE despite training on a smaller subset of the data.

	ANI-1 (Paper)	**This Work**
Dataset	GDB s01–s08, 17.2M conf.	GDB s01–s04 (subset)
Architecture (per element)	768:128:128:64:1	384:256:128:1
Parameters (per element)	124,033	131,585
Activation	Gaussian	ReLU

Optimizer / Loss	ADAM (exponential)	ADAM (MSE)
Reported metric	RMSE	MAE
Train error	1.2 kcal/mol	—
Validation error	1.3 kcal/mol	—
Test error	1.3 kcal/mol	—
Best single-run Test MAE	—	1.08 kcal/mol
5-Fold CV Test MAE	—	1.336 ± 0.213 kcal/mol
Multi-run Mean Test MAE	—	1.890 ± 0.353 kcal/mol
Final model (all data)	~1.3 kcal/mol (RMSE)	1.11 kcal/mol (MAE)

Table 2: Comparison with the ANI-1 paper. The original paper uses RMSE while this study uses MAE values.

Conclusion

In conclusion, the study demonstrates that an ANI-style neural network potential can achieve similar accuracy with the original ANI-1 paper even when trained on a smaller subset of data (GDB s01–s04). The best systematic test MAE of 1.24 kcal/mol (Model 3) and the final all-data model MAE of 1.11 kcal/mol are both comparative with the paper's reported test RMSE of 1.3 kcal/mol, considering the RMSE ≥ MAE characteristic. The 5-fold cross validation mean of 1.336 ± 0.213 kcal/mol is a robust estimate of generalization performance and confirms that the model consistently falls below the 2 kcal/mol accuracy target.

Architecture depth was the most important hyperparameter. AtomicNet_B, using two hidden layers (384 → 256 → 128 → 1), consistently outperformed AtomicNet_A that used a single hidden layer (384 → 128 → 1) with the best AtomicNet_B test MAE being 1.24 kcal/mol and the best AtomicNet_A test MAE being 1.76 kcal/mol. This follows the ANI-1 paper's finding that 3–4 hidden layer networks perform best. The closeness of the train, validation, and test MAEs for Model 3 confirm that the model generalizes well without overfitting. L2 regularization and lower learning rates did not improve performance at this data scale, and no dropout was found to be necessary.

The main limitation of this study is the variance (std = 0.353 kcal/mol) between runs. The full ANI-1 dataset of 17.2M conformations constrains the optimization landscape significantly more than our s01–s04 subset, leading to more stable convergence. The 5-fold CV result is more representative than any single run, and the all-data model is the most stable since it leverages the available data to the maximum.

Note that there is an architectural difference in that ANI-1 uses Gaussian activations, while this study uses ReLU. It is known that Gaussian activations produce smoother potential energy surfaces, which matters for molecular dynamics force calculations. Future work could explore CELU activations (used in torchani's reference implementation) or Gaussian activations to improve surface smoothness, training on the full s01–s08 dataset to reduce variance and make MAE closer to the 1 kcal/mol threshold for chemical accuracy, and implement a learning rate

schedule like the original paper had done. Overall, this project confirms that the AEV-based ANI architecture is effective and transferable even at reduced data scale.

References

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